

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Analyzed and mitigated patient safety hazards in software for cardiac medical products, achieving a **20% reduction** in identified risks through cross-functional collaboration with development and quality teams.
- Developed and maintained comprehensive risk management files in compliance with **ISO 13485** and **ISO 14971**, ensuring strict alignment with design inputs and outputs for medical device software.
- Executed reliability and safety engineering studies, including **design of experiments (DOE)**, to evaluate design concepts and system performance, leading to a **15% improvement** in product reliability metrics.
- Managed the traceability of design verification and validation activities within risk management documentation, ensuring all safety measures met **global regulatory standards** for implantable devices.
- Supported the design, maintenance, and continuation engineering of released software systems, including mobile applications, by conducting detailed **hazard analysis and risk analysis** techniques like PHA and DFMEA.
- Provided data-driven risk-benefit insights and proposed **innovative risk mitigation strategies** to stakeholders, prioritizing patient safety and contributing to a **10% decrease** in post-market surveillance findings.

SKILLS

Safety & Reliability Engineering – Risk Management, Hazard Analysis, Fault Tree Analysis (FTA), Failure Modes and Effects Analysis (FMEA), Design of Experiments (DOE), Reliability Engineering, Safety Engineering Studies, ISO 13485, ISO 14971, IEC 62304

Software Development – Java, Spring Boot, Python, .NET, C++, JavaScript, C#, Angular

Cloud & DevOps – AWS, Docker, Jenkins, Git, CI/CD

Data Analysis & BI – Power BI, Tableau, SQL, DAX, Power Query, R, pandas, NumPy

Methodologies & Practices – Agile, Scrum, Design for Reliability (DfR), Design for Six Sigma (DFSS), Change Management, Incident Response, Regulatory Compliance, GDPR Awareness

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Medical Device Software Safety & Reliability Enhancement Platform

- Architected and implemented a centralized platform for managing software safety and reliability risks for medical devices, integrating **ISO 13485** and **ISO 14971** compliance workflows.
- Developed automated pipelines using **Python** and **Jenkins** to analyze design verification outputs against risk management documentation, reducing manual review time by **30%** and ensuring traceability.

- Integrated **Splunk** for real-time monitoring of system performance and safety alerts, enabling proactive identification of potential hazards and supporting rapid incident response to maintain **99.95%** system uptime.
- Conducted comprehensive risk analyses, including PHA and DFMEA, on new software features and existing modules, resulting in the implementation of **15+ targeted risk mitigation strategies**.

Financial Services Transaction Monitoring Dashboard (HCL)

- Developed and maintained operational dashboards for a financial-services client, providing real-time transaction monitoring and alerting for failed payment messages.
- Automated data extraction and reporting processes using **Python** and **SQL** against PostgreSQL and Oracle databases, improving data availability for reconciliation workflows by **25%**.
- Managed production support activities, including P1/P2 incident response and SLA management, for a critical payments operations dashboard within a **GDPR-aware** environment.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)